

1. Implement Perceptron based IRIS classification.

CODE:

```
[13] ✓ 2s # Implement Perceptron based IRIS Classification
# Import Libraries
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load IRIS Dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split Dataset into Training and Testing Sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

# Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Create Perceptron Model
model = Perceptron(
    max_iter=1000,
    eta0=0.1,
    random_state=42
)

# Train Model
model.fit(X_train, y_train)
```

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[13] ✓ 2s # Predict Test Data
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy = {:.2f}%".format(accuracy * 100))

# Classification Report
print("\nClassification Report\n")
print(classification_report(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix\n")
print(cm)

# Plot Confusion Matrix
plt.figure(figsize=(7,6))
sns.heatmap(
    cm,
    annot=True,
    cmap="Blues",
    fmt="d",
    linewidths=1,
    linecolor="black",
    xticklabels=iris.target_names,
    yticklabels=iris.target_names
)

plt.title("Confusion Matrix - Perceptron IRIS Classification", fontsize=14)
plt.xlabel("Predicted Class", fontsize=12)
plt.ylabel("Actual Class", fontsize=12)
plt.show()
```

OUTPUT:

